
Measures of Dispersion

A decorative background consisting of a large number of small, semi-transparent dots in various colors including purple, teal, brown, and grey, arranged in a circular pattern that fades out towards the right side of the slide.

Definition :

A measure of scatter or spread of the data is called a measure of variation or measure of dispersion. A measure of dispersion give us idea about the reliability of averages.

- i. When dispersion is less the average is more reliable so it is a better estimate of population average.
- ii. When dispersion is more the average is not good representative of the data.



A	0 2 8 10 12 18 20
B	10 10 10 10 10 10 10 10
C	5 7 9 10 11 13 15

The following are some measures of dispersion

Absolute Measures (For comparison of same units) :

Range, Quartile Deviation or Semi-Interquartile Range , Mean Deviation ,
Standard Deviation

Relative Measures (For comparison of different units) :

Coefficient of Range , Coefficient of Quartile Deviation , Coefficient of Mean
Deviation, Coefficient of Variation



Requisite/Requirements of Good Measure of Dispersion

1. It should be easy to calculate and simple to understand
2. Should be rigidly defined
3. Should be based on all observations
4. Capable of further algebraic treatment
5. Should have sampling stability
6. Should be less affected by extreme values

Range:

$$\text{Range} = L - S$$

L : largest observation in the data

S : smallest observation in the data

Coefficient of Range:

$$\text{Coefficient of Range} = \frac{L-S}{L+S}$$

L : Largest observation in the data

S : Smallest observation in the data

**Compute Range and Coefficient of Range for the following data
115, 50, 120, 45, 68, 30, 170, 180, 65, 58**

$L = \text{Largest Observation} = 180$

$S = \text{Smallest Observation} = 30$

$$\begin{aligned}\text{Range} &= L - S \\ &= 180 - 30 \\ &= 150\end{aligned}$$

$$\begin{aligned}\text{Coeff of Range} &= \frac{L - S}{L + S} = \frac{180 - 30}{180 + 30} \\ &= \frac{150}{210} \\ &= 0.714286\end{aligned}$$

Find Range and Coefficient of Range for the following data

Petrol Consumption	100-150	150-300	300-450	450-600	600-750	750-1000
No of families	35	50	40	20	15	5

• $L = \text{Largest Obs} = 1000$
 $S = \text{Smallest Obs} = 100$

Range = $L - S$
 $= 1000 - 100$
 $= 900$

Coefficient of Range = $\frac{L - S}{L + S}$
 $= \frac{1000 - 100}{1000 + 100}$
 $= \frac{900}{1100}$
 $= 0.81818$

Quartile Deviation

Inter-Quartile Range : The Interquartile Range (IQR) formula is a measure of the middle 50% of a data set. The difference between the upper and lower quartile is known as the interquartile range.

$$\text{IQR} = Q_3 - Q_1$$

Semi Inter-Quartile Range/Quartile Deviation : Quartile deviation measures the spread or dispersion of data in a dataset. It is calculated as half of the difference between the first quartile (Q1) and the third quartile (Q3).

$$\text{QD} = \frac{Q_3 - Q_1}{2}$$

The corresponding relative measure of dispersion is called coefficient of quartile deviation and is defined as

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Calculate Quartile Deviation and Coefficient of Quartile Deviation for the following data

56,45,35,68,18,9,7,40,5,4,12,16,20,24,32

Arrange the data in ascending order

4, 5, 7, 9, 12, 16, 18, 20, 24, 32, 35, 40, 45, 56, 68

$$n = 15 \quad Q_i = i \left(\frac{n+1}{4} \right) \quad i = 1, 2, 3, 4$$

$$Q_1 = \left(\frac{n+1}{4} \right)^{\text{th}} \text{ observation}$$

$$= \left(\frac{15+1}{4} \right)^{\text{th}} \text{ observation}$$

$$= 4^{\text{th}} \text{ observation}$$

$$= 9$$

$$Q_3 = 3 \left(\frac{n+1}{4} \right)^{\text{th}} \text{ observation}$$

$$= 3 \left(\frac{16+1}{4} \right)^{\text{th}} \text{ observation}$$

$$= 12^{\text{th}} \text{ observation}$$

$$= 40$$

$$\text{Quartile Deviation} = \frac{Q_3 - Q_1}{2}$$

$$= \frac{40 - 9}{2}$$

$$= \frac{31}{2}$$

$$= 15.5$$

$$\text{Coeff of } QD = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

$$= \frac{40 - 9}{40 + 9}$$

$$= \frac{31}{49}$$

$$= 0.632653$$



Calculate Quartile Deviation and Coefficient of Quartile Deviation for the following data

Age	20-30	30-40	40-50	50-60	60-70
No of Person	35	45	30	27	13

$Q1 = 30.5556$

$Q3 = 50.92593$

$QD = 10.18519$

$\text{Coefficient of QD} = 0.25$

Age	No of Person	Less than type cf
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20-30	35	35
30-40	45	80
40-50	30	110
50-60	27	137
60-70	13	150
	<u>N=150</u>	

$$Q_1 = .$$

For Q_1

$$\frac{N}{4} = \frac{150}{4} = 37.5$$

Less than type cf ≥ 37.5 is 80

The quartile class Q_1 is 30-40

$$l_1 = 30 \quad l_2 = 40 \quad f = 45 \quad cf = 35$$

$$Q_1 = l_1 + \frac{(Q_2 - l_1) \left(\frac{N}{4} - cf \right)}{f}$$

$$\begin{aligned}
 &= 30 + (40 - 30) \left(\frac{37.5 - 35}{45} \right) \\
 &= 30 + (10) \left(\frac{2.5}{45} \right) \\
 &= 30.5556
 \end{aligned}$$

For Q_3

$$\frac{3N}{4} = \frac{3 \times 150}{4} = 112.5$$

Less than type cf ≥ 112.5 is 137

The quartile class Q_3 is 50-60

$$l_1 = 50 \quad l_2 = 60 \quad f = 27 \quad cf = 110$$

$$\begin{aligned}
 Q_3 &= l_1 + \frac{(l_2 - l_1)(\frac{3N}{4} - cf)}{f} \\
 &= 50 + \frac{(60 - 50)(112.5 - 110)}{27} \\
 &= 50 + (10) \left(\frac{2.5}{27} \right) \\
 &= 50.92593
 \end{aligned}$$

$$\begin{aligned}
 \text{Quartile Deviation} &= \frac{Q_3 - Q_1}{2} \\
 &= \frac{50.92593 - 30.5556}{2} \\
 &= 10.185165
 \end{aligned}$$

$$\begin{aligned}
 \text{Coefficient of QD} &= \frac{Q_3 - Q_1}{Q_3 + Q_1} \\
 &= \frac{50.92593 - 30.5556}{50.92593 + 30.5556} \\
 &= 0.25
 \end{aligned}$$

Calculate Quartile Deviation and Coefficient of Quartile Deviation for the following data

Age	20	30	40	50	60	70	80
No of Person	3	61	132	153	140	51	3

Age (in yrs)	No of Person	Less-than type cf
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20	3	3
30	61	64
40	132	196
50	153	349
60	140	489
70	51	540
80	3	543

$$N = 543$$

For Q_1

$$\left(\frac{N+1}{4}\right) = \left(\frac{543+1}{4}\right) = 136$$

Less than type cf ≥ 136 is 196

The corresponding value of age is 40.

$$\therefore Q_1 = 40 \text{ years}$$

For Q_3

$$3\left(\frac{N+1}{4}\right) = 3\left(\frac{543+1}{4}\right) = 408$$

Less than type cf ≥ 408 is 60

The corresponding value of age is 60

$$\therefore Q_3 = 60 \text{ years}$$

$$QD = \frac{Q_3 - Q_1}{2} = \frac{60 - 40}{2} = 10 \text{ years}$$

$$\text{coeff of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{60 - 40}{60 + 40} = \frac{20}{100} = 0.20$$

The scores of 30 students in each of 2 divisions A and B are given

Score	45-49	50-54	55-59	60-64	65-69	70-74	75-79	80-84
No of students								
DIV A	2	5	7	12	5	2	2	1
DIV B	1	3	4	10	12	4	1	1

Find coefficient of deviation for two divisions and comment

The given frequency distributions, after making the class-intervals continuous, can be written as :

Score	Div. A		Div. B	
	No. of Students	c.f. < type	No. of Students	c.f. < type
44.5-49.5	2	2	1	1
49.5-54.5	5	7	3	4
54.5-59.5	7	14	4	8
59.5-64.5	12	26	10	18
64.5-69.5	5	31	12	30
69.5-74.5	2	33	4	34
74.5-79.5	2	35	1	35
79.5-84.5	1	36	1	36

$$N = 36$$

For Div. A :

Calculation of Q_1 :

$$\frac{N}{4} = \frac{36}{4} = 9$$

Measures of

$\therefore Q_1$ class is (54.5 - 59.5)

$\therefore l_1 = 54.5, l_2 = 59.5, f = 7, c.f. = 7$

$$\therefore Q_1 = l_1 + \frac{(l_2 - l_1) \left(\frac{N}{4} - c.f. \right)}{f} = 54.5 + \frac{(59.5 - 54.5)(9 - 7)}{7}$$

$$= 54.5 + 1.43 = 55.93 \text{ units}$$

Calculation of Q_3 :

$$\frac{3N}{4} = \frac{3 \times 36}{4} = 27$$

$\therefore Q_3$ class is (64.5 - 69.5)

$\therefore l_1 = 64.5, l_2 = 69.5, f = 5, c.f. = 26$

$$\therefore Q_3 = l_1 + \frac{(l_2 - l_1) \left(\frac{3N}{4} - c.f. \right)}{f}$$

$$= 64.5 + \frac{(69.5 - 64.5)(27 - 26)}{5} = 64.5 + 1$$

$$= 65.5 \text{ units}$$

$$\therefore \text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{65.5 - 55.93}{65.5 + 55.93} = \frac{9.57}{121.43} = 0.0788.$$

For Div. B :

Calculation of Q_1 :

$$\frac{N}{4} = \frac{36}{4} = 9$$

$\therefore Q_1$ class is (59.5 - 64.5)

$\therefore l_1 = 59.5, l_2 = 64.5, f = 10, c.f. = 8$

$$\therefore Q_1 = l_1 + \frac{(l_2 - l_1) \left(\frac{N}{4} - c.f. \right)}{f}$$

$$= 59.5 + \frac{(64.5 - 59.5)(9 - 8)}{10}$$

$$= 59.5 + 0.5 = 60 \text{ units}$$

Calculation of Q_3 :

$$\frac{3N}{4} = \frac{3 \times 36}{4} = 27$$

$\therefore Q_3$ class is (64.5 - 69.5)

$\therefore l_1 = 64.5, l_2 = 69.5, f = 12, c.f. = 18$

$$\therefore Q_3 = l_1 + \frac{(l_2 - l_1) \left(\frac{3N}{4} - c.f. \right)}{f}$$

$$= 64.5 + \frac{(69.5 - 64.5)(27 - 18)}{12} = 64.5 + 3.75$$

$$= 68.25 \text{ units}$$

$$\therefore \text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{68.25 - 60}{68.25 + 60} = \frac{8.25}{128.25} = 0.0643.$$

$\therefore$ Coefficient of Q.D. is more for Div. A,

$\therefore$ There is more variation of marks in Div. A than in Div. B.

Mean Absolute Deviation or Mean Deviation

Mean Absolute Deviation (MAD) of a data set is the average distance between each data point of the data set and the mean of data. i.e. it represents the amount of variation that occurs around the mean value in the data set

Mean Deviation is defined as A.M. of absolute deviation of the observations from any constant value say “A”

$$\text{M.D from A} = \frac{1}{n} \sum |x-A|$$

For a frequency distribution where class intervals and frequencies are given

$$\text{M.D from A} = \frac{1}{N} \sum f |x-A| \text{ where } x_i \text{ and } f_i \text{ are class-mark and frequency of } i^{\text{th}} \text{ class } N = \sum f$$

The corresponding relative measure of dispersion is

$$\text{Coefficient of M.D. from A} = \frac{\text{MD from A}}{A}$$

Calculate M.D from mean and corresponding coefficient of M.D for the data

30,39,40,42,38,45,45,43,52,26

$$n = 10$$

$$\text{Mean } \bar{X} = \frac{\sum X}{n} = \frac{400}{10} = 40$$

$$\text{Mean Deviation From Mean} = \frac{\sum |X - \bar{X}|}{n}$$

X	$ X - \bar{X} = X - 40 $
30	10
39	1
40	0
42	2
38	2
45	5
45	5
43	3
52	12
26	14

$$\text{MD } \sum |X - \bar{X}| = 54$$

$$\text{MD from Mean} = \frac{\sum |X - \bar{X}|}{n} = \frac{54}{10} = 5.4$$

$$= \frac{\sum |X - 40|}{10}$$

$$\text{Coeff of MD} = \frac{\text{MD from Mean}}{\text{Mean}}$$

$$= \frac{5.4}{40}$$

$$= 0.135$$

Calculate mean deviation about

i. Mean

ii. Median

and corresponding relative measures for the following data

Class	20-40	40-60	60-80	80-100
Frequency	30	40	20	10

i. Mean deviation about mean and corresponding relative measures

Class	Frequency (f)	Class Mark (x)	fx	$ x - \text{Mean} $ $= x - 52 $	$f x - \text{Mean} $ $= f x - 52 $
20-40	30	30	900	22	660
40-60	40	50	2000	2	80
60-80	20	70	1400	18	360
80-100	10	90	900	38	380
	$\Sigma f = 100$		$\Sigma fx = 5200$		$\Sigma f x - 52 = 1480$

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f} = \frac{5200}{100} = 52$$

$$\text{MD about Mean} = \frac{\Sigma f|x - \text{Mean}|}{\Sigma f}$$

$$= \frac{1480}{100}$$

$$= 14.80$$

$$\text{Coeff of MD about mean} = \frac{\text{MD about Mean}}{\text{Mean}}$$

$$= \frac{14.80}{52}$$

$$= 0.2846$$

Mean deviation about median and corresponding relative measures

class	Frequency f	Less than type cf	class Mark(x)	$ x - \text{Median} $ $= x - 50 $	$f x - \text{Median} $
20-40	30	30	30	20	600
40-60	40	70	50	0	0
60-80	20	90	70	20	400
80-100	10	100	90	40	400
$\Sigma f = N = 100$					$\Sigma f x - \text{Median} = 1400$

$$\frac{N}{2} = \frac{100}{2} = 50$$

Less than type cf ≥ 50 is 70

The median class is 40-60

$$l_1 = 40 \quad l_2 = 60 \quad f = 40 \quad cf = 30$$

$$\text{Median} = l_1 + \frac{(l_2 - l_1) \left(\frac{N}{2} - cf \right)}{f}$$

$$= 40 + \frac{(60 - 40)(50 - 30)}{40}$$

$$= 40 + \frac{(20)(20)}{40}$$

$$= 50$$

Mean Deviation About Median

$$= \frac{\Sigma f|x - \text{Median}|}{\Sigma f}$$

$$= \frac{1400}{100} = 14$$

Coeff of Mean Deviation About Median

$$= \frac{\text{MD about Median}}{\text{Median}}$$

$$= \frac{14}{50}$$

$$= 0.28$$

Variance ,Standard Deviation and Coefficient of Variation

Variance : The A.M of squares of deviation taken from A.M is called variance denoted by σ^2 .

Standard Deviation : Square Root of Variance denoted by σ

Coefficient of Variation : Relative measure corresponding to standard deviation is Coefficient of Variation

$$CV = \frac{\sigma}{\bar{X}} \times 100$$

VARIANCE

Definition:

The variance is a measure of how far a set of data are dispersed out from their mean or average value. It is denoted as ' σ^2 '.

Properties of Variance:

It is always non-negative since each term in the variance sum is squared and therefore the result is either positive or zero.

Variance always has squared units. For example, the variance of a set of weights estimated in kilograms will be given in kg squared. Since the population variance is squared, we cannot compare it directly with the mean or the data themselves.

STANDARD DEVIATION

Definition:

The spread of statistical data is measured by the standard deviation. Distribution measures the deviation of data from its mean or average position. The degree of dispersion is computed by the method of estimating the deviation of data points. It is denoted by the symbol, ' σ '.

Properties of Standard Deviation

- It describes the square root of the mean of the squares of all values in a data set and is also called the root-mean-square deviation.
- The smallest value of the standard deviation is 0 since it cannot be negative.
- When the data values of a group are similar, then the standard deviation will be very low or close to zero. But when the data values vary with each other, then the standard variation is high or far from zero.

Merits:

1. It is rigidly defined and its value is always definite and based on all the observations and the actual signs of deviations are used.
2. As it is based on arithmetic mean, it has all the merits of arithmetic mean.
3. It is the most important and widely used measure of dispersion.
4. It is possible for further algebraic treatment.
5. It is less affected by the fluctuations of sampling and hence stable.
6. It is the basis for measuring the coefficient of correlation and sampling.

Demerits:

1. It is not easy to understand and it is difficult to calculate.
2. It gives more weight to extreme values because the values are squared up.
3. As it is an absolute measure of variability, it cannot be used for the purpose of comparison.

VARIANCE AND STANDARD DEVIATION FORMULA

Variance for Raw Data

$$\text{VARIANCE} = \frac{\sum x^2}{n} - \bar{x}^2$$

x_i : i th observation in the population

n : no of observations

$\bar{x}$

Variance for Frequency distribution

$$\text{Variance, } \sigma^2 = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2$$

x : Values of each observation

f : Frequency corresponding to each value of x

Standard deviation is square root of variance $= \sqrt{\text{Variance}}$

Calculate S.D for the following data on age of 7 students

17,18,20,17,20,21,19

X	X ²
17	(17) ² =289
18	(18) ² =324
20	400
17	289
20	400
21	441
19	361

$$\sum X = 132 \quad \sum X^2 = 2504$$

$$n = 7$$

$$\bar{X} = \frac{\sum X}{n} = \frac{132}{7} = 18.8571$$

$$S.D = \sqrt{\frac{1}{n} \sum X^2 - \bar{X}^2}$$

$$= \sqrt{\frac{2504}{7} - (18.8571)^2}$$

$$= 1.4574 \text{ years}$$

Find S.D for the age-distribution of 100 students:

Age(x)(in years)	17	18	19	20	21	22	23
No of Students(f)	7	11	17	30	15	12	8

Solution :

x	f	fx	fx^2
17	7	119	2023
18	11	198	3564
19	17	323	6137
20	30	600	12000
21	15	315	6615
22	12	264	5808
23	8	184	4232
Total	100	2003	40379

$$\Sigma f = 100, \Sigma fx = 2003, \Sigma fx^2 = 40379$$

$$\therefore \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{2003}{100} = 20.03$$

$$\begin{aligned} \text{S.D.}(x) &= \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2} = \sqrt{\frac{40379}{100} - (20.03)^2} \\ &= \sqrt{2.5891} \\ &= 1.6091 \text{ years.} \end{aligned}$$

Calculate average and S.D of profit (in lakhs) for different branches of a company

Profit(in lakhs)	10-20	20-30	30-40	40-50	50-60
No of branches	8	12	20	6	4

Solution :

Profit	No. of Branches f	Class-mark x	fx	fx^2
10-20	8	15	120	1800
20-30	12	25	300	7500
30-40	20	35	700	24500
40-50	6	45	270	12150
50-60	4	55	220	12100
	50		1610	58050

$$\Sigma f = 50, \Sigma fx = 1610, \Sigma fx^2 = 58050$$

$$\therefore \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{1,610}{50} = ₹ 32.20 \text{ crores}$$

$$\text{S.D.}(x) = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2} = \sqrt{\frac{58050}{50} - (32.20)^2}$$

$$= \sqrt{124.19} = ₹ 11.1427 \text{ lakhs}$$

Combined Variance

Suppose we have two sets of data containing n_1 and n_2 observations with means $\overline{X_1}$ and $\overline{X_2}$ and variances S_1^2 and S_2^2 . If $\overline{X_c}$ is the combined mean and S_c^2 is the combined variance of $(n_1 + n_2)$ observations, then combined variance is given by:

$$S_c^2 = \frac{n_1 S_1^2 + n_2 S_2^2 + n_1 (\overline{X_1} - \overline{X_c})^2 + n_2 (\overline{X_2} - \overline{X_c})^2}{n_1 + n_2}$$

The above equation can be further solved and re-written as;

$$S_c^2 = \frac{n_1 [S_1^2 + (\overline{X_1} - \overline{X_c})^2] + n_2 [S_2^2 + (\overline{X_2} - \overline{X_c})^2]}{n_1 + n_2}$$

$$\text{Where } \overline{X_c} = \frac{n_1 * \overline{X_1} + n_2 * \overline{X_2}}{n_1 + n_2}$$

The combine standard deviation S_c can be calculated by taking square root of S_c^2

$$S_c^2 = \frac{n_1 S_1^2 + n_2 S_2^2 + n_1 (\overline{X_1} - \overline{X_c})^2 + n_2 (\overline{X_2} - \overline{X_c})^2}{n_1 + n_2}$$

The above equation can be further solved and re-written as;

$$S_c^2 = \frac{n_1 [S_1^2 + (\overline{X_1} - \overline{X_c})^2] + n_2 [S_2^2 + (\overline{X_2} - \overline{X_c})^2]}{n_1 + n_2}$$

Example :For a group of 50 male workers the mean and standard deviation of their daily wages are 63 dollars and 9 dollars respectively. For a group of 40 female workers these values are 54 dollars and 6 dollars respectively. Find the mean and variance of the combined group of 90 workers.

Solution:

Here; $n_1 = 50$; $\overline{X_1} = 63$ and $S_1^2 = 81$

$n_2 = 40$; $\overline{X_2} = 54$ and $S_2^2 = 36$

Combine arithmetic mean $\overline{X_c} = \frac{n_1 * \overline{X_1} + n_2 * \overline{X_2}}{n_1 + n_2} = \frac{50(63) + 40(54)}{50 + 40} = \frac{5310}{90}$

$\overline{X_c} = 59$

Combine Variance $S_c^2 = \frac{n_1[S_1^2 + (\overline{X_1} - \overline{X_c})^2] + n_2[S_2^2 + (\overline{X_2} - \overline{X_c})^2]}{n_1 + n_2}$

$$S_c^2 = \frac{50[81 + (63 - 59)^2] + 40[36 + (54 - 59)^2]}{50 + 40} = \frac{7290}{90} = 81$$

A group of 50 observations has A.M 61 and S.D 8 . Another group of 100 observations has A.M. 70 and S.D 9. Find mean & S.D for the combined group of 150 observations

S.D= 9.6609

The number of runs scored by two cricketers A and B in 10 innings of 5 test matches below

Find which cricketer is

i. Better in average

ii. More consistent

are shown below :

<i>A</i>	5	20	90	76	102	90	6	108	20	16
<i>B</i>	40	35	60	62	58	76	42	30	30	20

Solution :

$$\text{Mean of } A = \bar{x}_A = \frac{\Sigma x_A}{n} = \frac{533}{10} = 53.3$$

$$\text{S.D.}(A) = \sqrt{\frac{\Sigma x_A^2}{n} - \bar{x}_A^2} = \sqrt{\frac{45161}{10} - (53.3)^2} = 40.9293$$

$$\therefore (\text{C.V.})_A = \frac{\text{S.D.}(A)}{\bar{x}_A} \times 100 = \frac{40.9293}{53.3} \times 100 = 76.79\%$$

$$\text{Mean of } B = \bar{x}_B = \frac{\Sigma x_B}{n} = \frac{453}{10} = 45.3$$

$$\text{S.D.}(B) = \sqrt{\frac{\Sigma x_B^2}{n} - \bar{x}_B^2} = \sqrt{\frac{23373}{10} - (45.3)^2} = 16.8882$$

$$\therefore (\text{C.V.})_B = \frac{\text{S.D.}(B)}{\bar{x}_B} \times 100 = \frac{16.8882}{45.3} \times 100 = 37.28\%$$

(i) $\bar{x}_A > \bar{x}_B$, cricketer A is better in average.

(ii) $(\text{C.V.})_B < (\text{C.V.})_A$, cricketer B is more consistent.

Moments

- Central tendency- Average of the set of data
- Dispersion- Spread of the data from average value
- Moments- Shape of the distribution.
- Moments are of two types-
 1. Raw moments- about origin (zero)
 2. Central Moments- about mean ($\bar{X}$)

Moments of Raw Data

Central Moments- Arithmetic mean of various powers of deviations taken from mean $\bar{X}$.

$$\mu_r = \frac{\sum (X - \bar{X})^r}{N}$$

Raw Moments- Arithmetic mean of various powers of deviations taken from the origin.

$$\mu'_r = \frac{\sum X^r}{N}$$

Moments of Grouped Data

Central Moments- Arithmetic mean of various powers of deviations taken from mean $\bar{X}$.

$$\mu_r = \frac{\sum f(X - \bar{X})^r}{\sum f}$$

Raw Moments- Arithmetic mean of various powers of deviations taken from the origin.

$$\mu'_r = \frac{\sum fX^r}{\sum f}$$

Relation between raw moments and central moments

$$\mu'_1 = \bar{X} = \text{Arithmetic Mean}$$

$$\mu_1 = 0 \quad (\text{When taken about mean})$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2 \quad (\text{Variance})$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3$$

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6(\mu'_1)^2\mu'_2 - 3(\mu'_1)^4$$

Note :

- The first Raw moment is the arithmetic mean
- The first central moment when taken ab

Find the (i) first, (ii) second, (iii) third and (iv) fourth moments about the mean of the set 2, 3, 7, 8, 10.

The given data is 2, 3, 7, 8, 10

Moments about mean

X	$(X - \bar{X})$ $= (X - 6)$	$(X - \bar{X})^2$ $= (X - 6)^2$	$(X - \bar{X})^3$ $= (X - 6)^3$	$(X - \bar{X})^4$ $= (X - 6)^4$
2	-4	16	-64	256
3	-3	9	-27	81
7	1	1	1	1
8	2	4	8	16
10	4	16	64	256
$\Sigma X = 30$	$\Sigma (X - \bar{X}) = 0$	$\Sigma (X - \bar{X})^2 = 46$	$\Sigma (X - \bar{X})^3 = -18$	$\Sigma (X - \bar{X})^4 = 610$

$$\bar{X} = \frac{\Sigma X}{n} = \frac{30}{5} = 6$$

$$\mu_r = \frac{\Sigma (X - \bar{X})^r}{n}$$

$$\mu_1 = \frac{\Sigma (X - \bar{X})}{n}$$

$$= 0$$

$$\mu_2 = \frac{\Sigma (X - \bar{X})^2}{n}$$

$$= \frac{46}{5}$$

$$= 9.2$$

$$\mu_3 = \frac{\Sigma (X - \bar{X})^3}{n}$$

$$= \frac{-18}{5}$$

$$= -3.6$$

$$\mu_4 = \frac{\Sigma (X - \bar{X})^4}{n}$$

$$= \frac{610}{5}$$

$$\mu_4 = 122$$

Example 2:

Find first four moments- 1, 2, 3, 3, 6

- i. About point 4**
- ii. About zero**
- iii. About mean**

i) About Point 4

x	(x-4)	(x-4) ²	(x-4) ³	(x-4) ⁴
1	-3	9	-27	81
2	-2	4	-8	16
3	-1	1	-1	1
3	-1	1	-1	1
6	2	4	8	16

$$\sum x = 15 \quad \sum (x-4) = -5 \quad \sum (x-4)^2 = 19 \quad \sum (x-4)^3 = -29 \quad \sum (x-4)^4 = 115$$

$$\mu_1 = \frac{\sum (x-4)^1}{n}$$

$$= \frac{-5}{5}$$

$$= -1$$

$$\mu_2 = \frac{\sum (x-4)^2}{n}$$

$$= \frac{19}{5}$$

$$= 3.8$$

$$\mu_3 = \frac{\sum (x-4)^3}{n}$$

$$= \frac{-29}{5} = -5.8$$

$$\mu_4 = \frac{\sum (x-4)^4}{n}$$

$$= \frac{115}{5} = 23$$

ii) About zero (Raw moments)

x	x ²	x ³	x ⁴
1	1	1	1
2	4	8	16
3	9	27	81
3	9	27	81
6	36	216	1296

$$\sum x = 15 \quad \sum x^2 = 59 \quad \sum x^3 = 279 \quad \sum x^4 = 1475$$

$$\mu_1' = \frac{\sum x^1}{n} = \frac{15}{5} = 3$$

$$\mu_2' = \frac{\sum x^2}{n} = \frac{59}{5} = 11.8$$

$$\mu_3' = \frac{\sum x^3}{n} = \frac{279}{5} = 55.8$$

$$\mu_4' = \frac{\sum x^4}{n} = \frac{1475}{5} = 295$$

About Mean

x	$(x - \bar{x})$	$(x - \bar{x})^2$	$(x - \bar{x})^3$	$(x - \bar{x})^4$
1	-2	4	-8	16
2	-1	1	-1	1
3	0	0	0	0
3	0	0	0	0
6	3	9	27	81
$\Sigma x = 15$ $\Sigma(x - \bar{x}) = 0$ $\Sigma(x - \bar{x})^2 = 14$ $\Sigma(x - \bar{x})^3 = 18$ $\Sigma(x - \bar{x})^4 = 98$				

$$\bar{x} = \frac{\Sigma x}{n} = \frac{15}{5} = 3$$

$$\mu_1 = \frac{\Sigma(x - \bar{x})}{n} = 0$$

$$\mu_2 = \frac{\Sigma(x - \bar{x})^2}{n} = \frac{14}{5} = 2.8$$

$$\mu_3 = \frac{\Sigma(x - \bar{x})^3}{n} = \frac{18}{5} = 3.6$$

$$\mu_4 = \frac{\Sigma(x - \bar{x})^4}{n} = \frac{98}{5} = 19.6$$

1. Calculate first four moments about zero and about mean.

Class	1 - 3	3 - 5	5 - 7	7 - 9	9 - 11	11 - 13	13 - 15
Frequency	1	3	5	8	4	3	1

①	②	③	④	⑤	⑥	⑦	⑧	⑨
Class	Freq f	Class Mark x	fx	(x- $\bar{x}$)	f(x- $\bar{x}$) = ② x ⑤	f(x- $\bar{x}$) ² = ② x ⑥	f(x- $\bar{x}$) ³ = ② x ⑦	f(x- $\bar{x}$) ⁴ = ② x ⑧
1-3	1	2	2	-5.92	1x(-5.92) =-5.92	1x(-5.92) ² =35.0464	1x(-5.92) ³ =-207.0714	1x(-5.92) ⁴ =1228.24
3-5	3	4	12	-3.92	3x(-3.92) =-11.76	3x(-3.92) ² =46.0932	3x(-3.92) ³ =-180.7038	3x(-3.92) ⁴ =708.328
5-7	5	6	30	-1.92	5x(-1.92) =-9.6	5x(-1.92) ² =18.432	5x(-1.92) ³ =-35.3894	5x(-1.92) ⁴ =67.947
7-9	8	8	64	0.08	8x(0.08) =0.64	8x(0.08) ² =0.0512	8x(0.08) ³ =0.004096	8x(0.08) ⁴ =0.00032
9-11	4	10	40	2.08	4x(2.08) =8.32	4x(2.08) ² =17.3056	4x(2.08) ³ =35.9956	4x(2.08) ⁴ =74.8706
11-13	3	12	36	4.08	3x(4.08) =12.24	3x(4.08) ² =49.9392	3x(4.08) ³ =203.7519	3x(4.08) ⁴ =831.307
13-15	1	14	14	6.08	1x(6.08) =6.08	1x(6.08) ² =36.9664	1x(6.08) ³ =224.7551	1x(6.08) ⁴ =1366.51
	$\Sigma f = 25$		$\Sigma fx = 198$	$\Sigma (x - \bar{x}) = 0$	$\Sigma f(x - \bar{x}) = 0$	$\Sigma f(x - \bar{x})^2 = 203.84$	$\Sigma f(x - \bar{x})^3 = 40.934$	$\Sigma f(x - \bar{x})^4 = 4277.271$

Four moments about mean

$$\text{Mean } \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{198}{25} = 7.92$$

$$\mu_1 = \frac{\Sigma f(x - \bar{x})}{\Sigma f} = 0$$

$$\mu_2 = \frac{\Sigma f(x - \bar{x})^2}{\Sigma f} = \frac{203.84}{25} = 8.1536$$

$$\mu_3 = \frac{\Sigma f(x - \bar{x})^3}{\Sigma f} = \frac{40.934}{25} = 1.63736$$

$$\mu_4 = \frac{\Sigma f(x - \bar{x})^4}{\Sigma f} = \frac{4277.271}{25}$$

$$= 171.0908$$

2. Calculate first and second moments about mean

Size	0-4	4-8	8-12	12-16
Frequency	10	12	15	13

① Size	② freq f	③ Class Mark(x)	④ fx	⑤ (x - $\bar{x}$)	⑥ f(x - $\bar{x}$) = ② x ⑤	⑦ f(x - $\bar{x}$) ² = ② x ⑤ ²
0-4	10	2	20	-6.48	-64.8	419.904
4-8	12	6	72	-2.48	-29.76	73.8048
8-12	15	10	150	1.52	22.8	34.656
12-16	13	14	182	5.52	71.76	396.1152
	$\Sigma f = 50$		$\Sigma fx = 424$	0	$\Sigma f(x - \bar{x}) = 0$	$\Sigma f(x - \bar{x})^2 = 924.48$

$$\text{Mean } \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{424}{50} = 8.48$$

Moments about mean

$$\mu_1 = \frac{\Sigma f(x - \bar{x})}{\Sigma f}$$

$$= 0$$

$$\mu_2 = \frac{\Sigma f(x - \bar{x})^2}{\Sigma f}$$

$$= \frac{924.48}{50}$$

$$= 18.4896$$

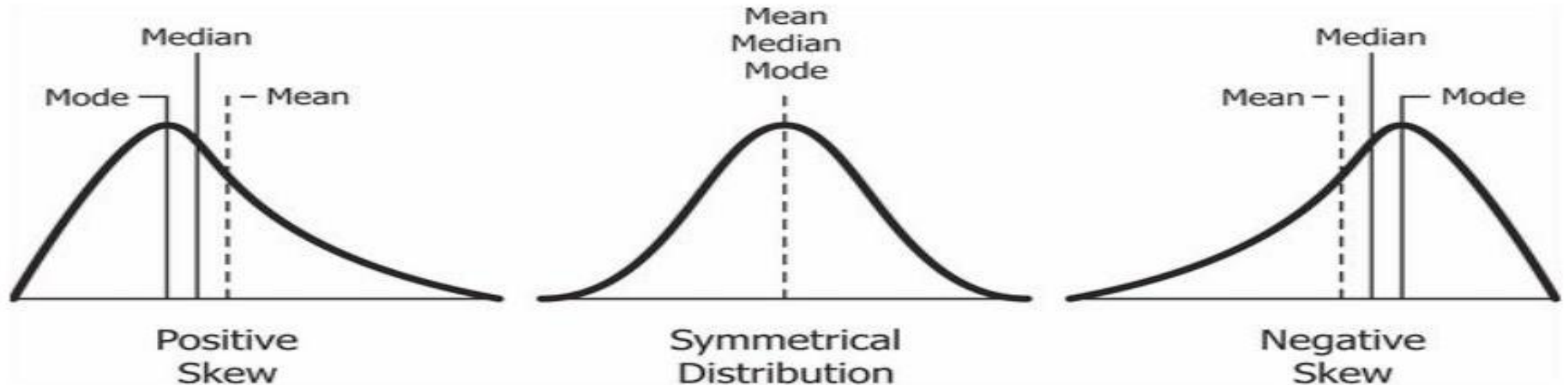
Skewness

Skewness is a degree of asymmetry/ lack of symmetry/ departure from symmetry.

Concerned with the shape of the curve .

Three types-

1. Positive Skewness
2. Symmetrical Distribution
3. Negative Skewness



Positive Skewness –

Means when the tail on the right side of the distribution is longer.

The mass of the distribution is concentrated on the left of the figure

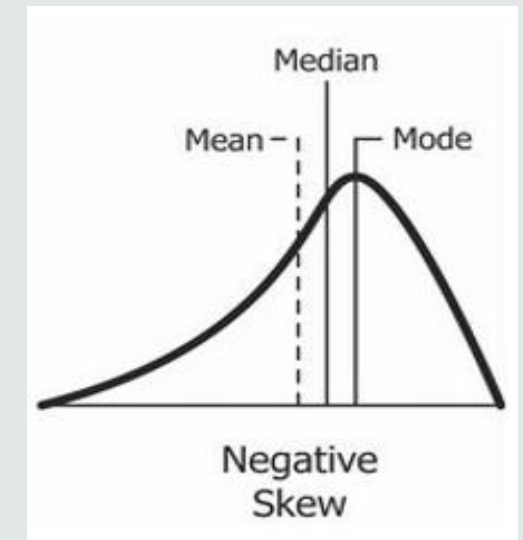
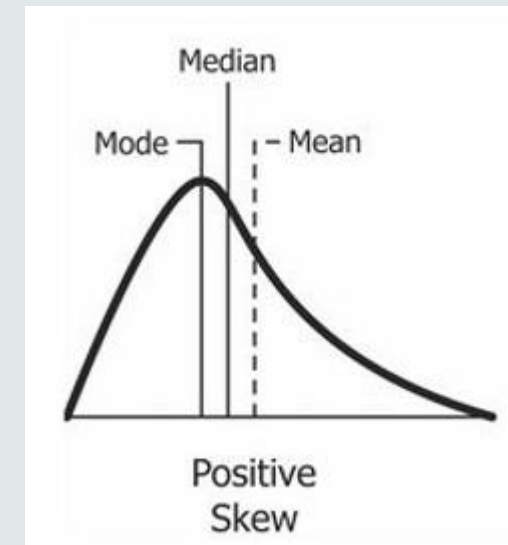
The mean and median will be greater than the mode.

Negative Skewness-

is when the tail of the left side of the distribution is longer.

the mass of the distribution is concentrated on the right of the figure

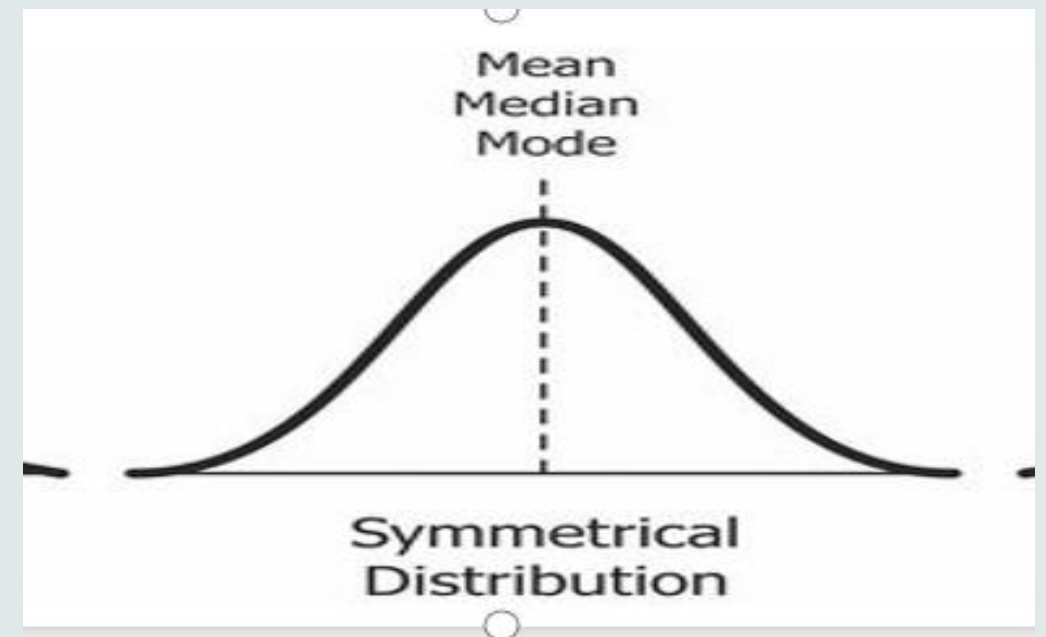
The mean and median will be less than the mode.



Symmetrical Distribution:

A distribution is symmetric if it looks same to the right and to the left of the center point

The mean median and mode will be same



KARL PEARSON COEFFICIENT OF SKEWENESS

Coefficient of skewness = SK_P

$$SK_P = \frac{\text{Mean} - \text{Mode}}{S.D} \quad \dots \text{Pearson's 1}^{\text{st}} \text{ coefficient of skewness}$$

$$SK_P = \frac{3(\text{Mean} - \text{Median})}{\sigma} \quad \dots \text{Pearson's 2}^{\text{nd}} \text{ coefficient of skewness}$$

The distribution is positively or negatively skewed depending on whether the value of SK_P is positive or negative

Example: Compute coefficient of skewness for the following observations
2, 3, 5, 7, 4, 8, 1.

The given data is 2, 3, 5, 7, 4, 8, 1

SKp
Karl Pearson Coeff of Skewness

$$SKp = \frac{\text{Mean} - \text{Mode}}{S.D(\sigma)} \quad \frac{3(\text{Mean} - \text{Median})}{\sigma}$$

Arrange the data in ascending order
1, 2, 3, 4, 5, 7, 8
Median = 4

$$\text{Mean} = \frac{\sum x_i}{n} = \frac{1+2+3+4+5+7+8}{7}$$
$$= \frac{30}{7}$$
$$= 4.2857$$
$$SKp = \frac{3(4.2857 - 4)}{\sigma} \quad \frac{30 - (1)^2 + (2)^2 + (3)^2 + (4)^2 + (5)^2 + (7)^2 + (8)^2}{n}$$
$$S.D(\sigma) = \sqrt{\frac{\sum x^2}{n} - (\bar{x})^2}$$
$$= \sqrt{\frac{168}{7} - (4.2857)^2}$$
$$(\sigma) = 2.3733$$
$$SKp = \frac{3(\text{Mean} - \text{Median})}{\sigma} = \frac{3(4.2857 - 4)}{2.3733}$$
$$= 0.36114$$

$SKp > 0$ The given data is positively skewed

Calculate Karl Pearson Coefficient of Skewness for the following data and comment on the answer

Class Interval	0-7	7-14	14-21	21-28	28-35	35-42	42-49	49-56
Frequency	26	31	35	42	82	71	54	19

Solution :

Class-interval	Class-mark x	Frequency f	fx	fx^2
0-7	3.5	26	91.0	318.50
7-14	10.5	31	325.5	3417.75
14-21	17.5	35	612.5	10718.75
21-28	24.5	42	1029.0	25210.50
28-35	31.5	82	2583.0	81364.50
35-42	38.5	71	2733.5	105239.75
42-49	45.5	54	2457.0	111793.50
49-56	52.5	19	997.5	52368.75
		$N = 360$	10829.0	390432

$$\text{Mean} = \bar{x} = \frac{\sum fx}{\sum f} = \frac{10829}{360} = 30.0806$$

$$\begin{aligned}\text{S.D.} &= \sqrt{\frac{\sum fx^2}{\sum f} - \bar{x}^2} = \sqrt{\frac{390432}{360} - (30.0806)^2} \\ &= \sqrt{179.690837} = 13.4049\end{aligned}$$

$\therefore$ Highest frequency is 82, $\therefore$ Modal class is (28-35)

$$\therefore l_1 = 28, l_2 = 35, f_0 = 82, f_1 = 42, f_2 = 71$$

$$\begin{aligned}\therefore \text{Mode} &= l_1 + \frac{(l_2 - l_1)(f_0 - f_1)}{(f_0 - f_1) + (f_0 - f_2)} = 28 + \frac{(35 - 28)(82 - 42)}{(82 - 42) + (82 - 71)} \\ &= 28 + 5.4902 \\ &= 33.4902\end{aligned}$$

$$\begin{aligned}\therefore \text{SK}_P &= \frac{\text{Mean} - \text{Mode}}{\text{S.D.}} = \frac{30.0806 - 33.4902}{13.4049} \\ &= -0.2544\end{aligned}$$

$\therefore$ Coefficient of skewness is negative, the given data is negatively skewed.

Bowleys Coefficient of Skewness

Bowleys Coefficient of Skewness= SK_B

$$SK_B = \frac{(Q3 - Median) - (Median - Q1)}{(Q3 - Q1)}$$

$$= \frac{Q3 - 2Median + Q1}{Q3 - Q1} = \frac{Q3 - 2Q2 + Q1}{Q3 - Q1}$$

If $SK_B > 0$ the distribution is positively skewed

If $SK_B < 0$ the distribution is negatively skewed

NOTE: Bowleys Coefficient of Skewness is useful when we have open end class intervals or unequal class intervals

Obtain coefficient of Skewness for the following data and comment on the nature of the distribution

Height(in inches)	Less than 61	61-63	63-65	65-67	67-69
No of Females	9	20	35	10	6

Solution :

Since the frequency distribution has open-end class-interval, we find Bowley's coefficient of skewness, SK_B .

Height (in inches)	No of females	Less than type c.f.
less than 61	9	9
61-63	20	29
63-65	35	64
65-67	10	74
67-69	6	80

$$\text{For } Q_1 : \frac{N}{4} = \frac{80}{4} = 20$$

$\therefore Q_1$ class is (61 - 63)

$$\therefore l_1 = 61, l_2 = 63, c.f. = 9, f = 20$$

$$\begin{aligned}\therefore Q_1 &= l_1 + \frac{(l_2 - l_1) \left(\frac{N}{4} - c.f. \right)}{f} \\ &= 61 + \frac{(63 - 61)(20 - 9)}{20} = 62.1\end{aligned}$$

$$\text{For } Q_2 : \frac{N}{2} = \frac{80}{2} = 40 \quad \therefore Q_2 \text{ class is (63 - 65)}$$

$$\therefore l_1 = 63, l_2 = 65, c.f. = 29, f = 35$$

$$\begin{aligned}\therefore Q_2 &= l_1 + \frac{(l_2 - l_1) \left(\frac{N}{2} - c.f. \right)}{f} \\ &= 63 + \frac{(65 - 63)(40 - 29)}{35} = 63.62857\end{aligned}$$

$$\text{For } Q_3 : \frac{3N}{4} = 60 \quad \therefore Q_3 \text{ class is (63 - 65) [same as } Q_2 \text{ class]}$$

$$\begin{aligned}\therefore Q_3 &= l_1 + \frac{(l_2 - l_1) \left(\frac{3N}{4} - c.f. \right)}{f} \\ &= 63 + \frac{(65 - 63)(60 - 29)}{35} = 64.7714\end{aligned}$$

$$\therefore SK_B = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} = -0.14438$$

Coefficient of skewness SK_B is negative, the distribution is negatively skewed

Relative Measures Based on Moments

$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$ and $\gamma_1 = \pm \sqrt{\beta_1}$ algebraic sign of γ_1 *being same as that of μ_3*

If $\gamma_1 = 0$ then the distribution is symmetric

If $\gamma_1 > 0$ then the distribution is positively skewed

If $\gamma_1 < 0$ then the distribution is negatively skewed

The first 3 moments about origin zero for the distribution are 2,25,80 respectively. Find measures of skewness based on central moments and comment on the answer

$$\text{Given : } \mu'_1 = 2, \mu'_2 = 25, \mu'_3 = 40$$

$$\mu_2 = \mu'_2 - \mu'^2_1 = 25 - 4 = 21$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu'^3_1 = 80 - 150 + 16 = -54$$

$$\therefore \beta_1 = \frac{\mu^2_3}{\mu^3_2} = \frac{(-54)^2}{(21)^3} = 0.314868$$

$$\gamma_1 = -\sqrt{\beta_1} = -\sqrt{0.314868} \quad (\because \mu_3 \text{ is negative}) \\ = -0.5611317$$

$\therefore \gamma_1 < 0$, the distribution is negatively skewed.

Kurtosis

The degree of “peakedness/flatness” of a distribution is known as Kurtosis. Usually it is taken compared to the normal distribution.

Leptokurtic- High peak- Lepto- Skinny peak- +ve kurtosis

Mesokurtic- Normal distribution curve- zero kurtosis

Platykurtic- Flat-topped- platy-broad peak- -ve kurtosis

Coefficient of Kurtosis

$$\gamma_2 = \beta_2 - 3 \quad \text{where } \beta_2 = \frac{\mu_4}{\mu_2^2}$$

If $\gamma_2 = 0$ i.e. $\beta_2 = 3$ the distribution is mesokurtic

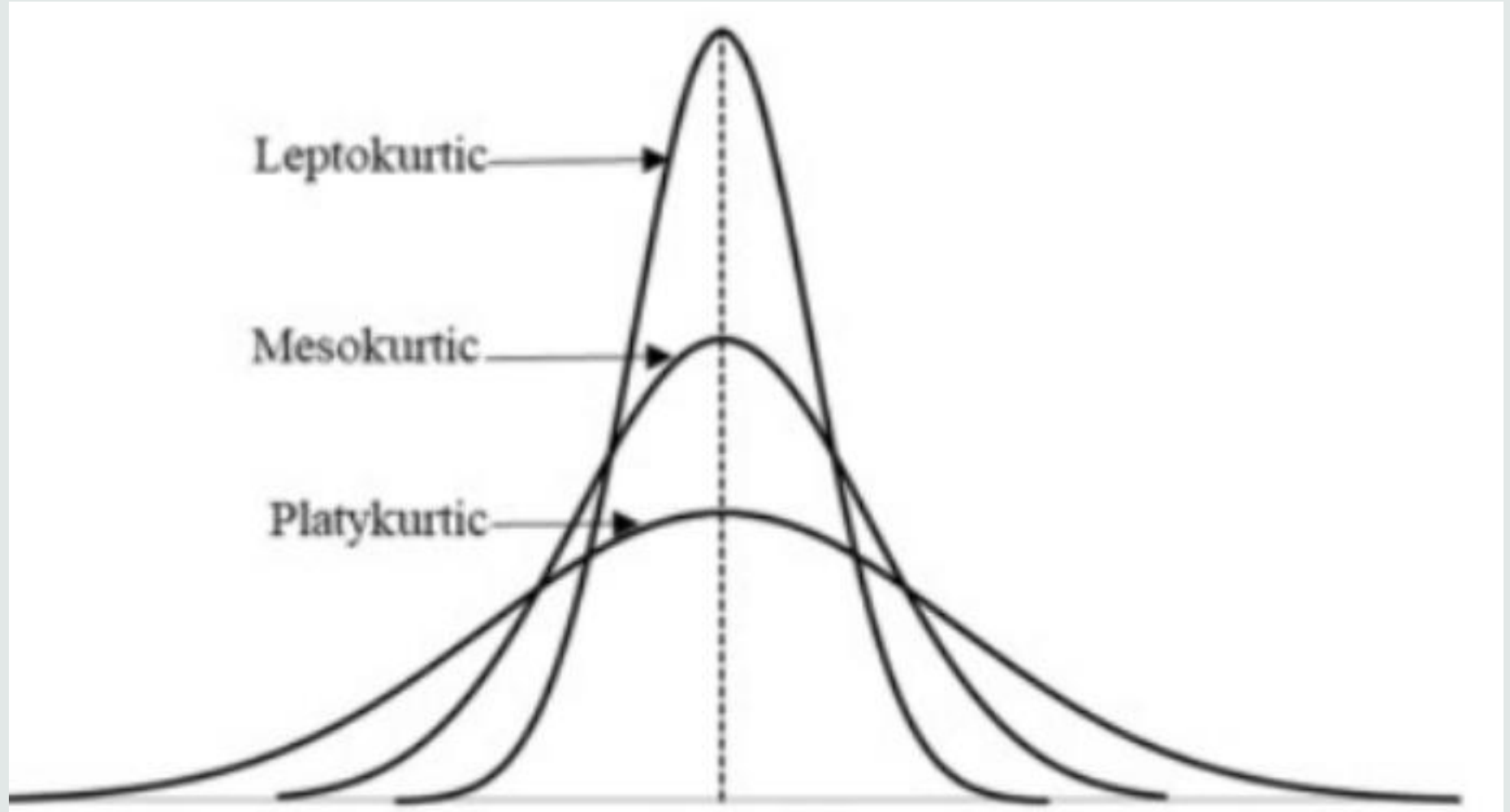
If $\gamma_2 > 0$ i.e. $\beta_2 > 3$ the distribution is leptokurtic

If $\gamma_2 < 0$ i.e. $\beta_2 < 3$ the distribution is platykurtic

Leptokurtic

Mesokurtic

Platykurtic



For the following data find β_2 and comment on the answer

$$\sum f = 100, \sum fx = 50, \sum fx^2 = 1967.2, \sum fx^3 = 2925.8, \sum fx^4 = 86650.2$$

Solution :

$$\mu_1' = \frac{\sum fx}{\sum f} = \frac{50}{100} = 0.5$$

$$\mu_2' = \frac{\sum fx^2}{\sum f} = \frac{1967.2}{100} = 19.672$$

$$\mu_3' = \frac{\sum fx^3}{\sum f} = \frac{2925.8}{100} = 29.258$$

$$\mu_4' = \frac{\sum fx^4}{\sum f} = \frac{86650.2}{100} = 866.502$$

$\therefore$ The central moments are

$$\mu_2 = \mu_2' - \mu_1'^2 = 19.672 - (0.5)^2 = 19.672 - 0.25 = 19.422$$

$$\begin{aligned}\mu_4 &= \mu_4' - 4\mu_1'\mu_3' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4 \\ &= 866.502 - 4 \times (0.5) (29.258) + 6 \times (0.5)^2 \times (19.672) - 3 \times (0.5)^4 \\ &= 866.502 - 58.516 + 29.508 - 0.1875 \\ &= 837.3065\end{aligned}$$

$$\therefore \beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{837.3065}{(19.422)^2} = \frac{837.3065}{377.214084} = 2.2197$$

$\therefore \beta_2 < 3$, the given distribution is platykurtic.

Find Measure of Kurtosis for the following data

x	1	2	3	4	5	6	7
f	5	38	65	92	70	40	10

x	f	fx	fx^2	fx^3	fx^4
1	5	5	5	5	5
2	38	76	152	304	608
3	65	195	585	1755	5265
4	92	368	1472	5888	23552
5	70	350	1750	8750	43750
6	40	240	1440	8640	51840
7	10	70	490	3430	24010
	320	1304	5894	28772	149030

$$\therefore \mu'_1 = \frac{\sum fx}{f} = \frac{1304}{320} = 4.075$$

$$\mu'_2 = \frac{\sum fx^2}{f} = \frac{5894}{320} = 18.41875$$

$$\mu'_3 = \frac{\sum fx^3}{f} = \frac{28772}{320} = 89.9125$$

$$\mu'_4 = \frac{\sum fx^4}{f} = \frac{149030}{320} = 465.71875$$

$$\therefore \mu_2 = \mu'_2 - \mu_1'^2 = 18.41875 - (4.075)^2 = 1.813125$$

$$\begin{aligned} \mu_4 &= \mu'_4 - 4\mu_1'\mu'_3 + 6\mu_1'^2\mu'_2 - 3\mu_1'^4 \\ &= 465.71875 - 4 \times (4.075) \times (89.9125) + 6 \\ &\quad \times (4.075)^2 \times (18.41875) - 3 \times (4.075)^4 \\ &= 465.71875 - 1465.57375 + 1835.129133 - 827.240345 \\ &= 8.033788 \end{aligned}$$

$$\therefore \beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{8.033788}{(1.813125)^2} = \frac{8.033788}{3.287422} = 2.4438$$

$$\therefore \gamma_2 = \beta_2 - 3 = -0.5562$$

Mean Squared Error or Mean Squared Deviation

It is defined as the average squared difference between the estimated values and the actual value.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

MSE = mean squared error

n = number of data points

Y_i = observed values

$\hat{Y}_i$ = predicted values

Effect of Change of Origin and Scale on Variance and S.D

1. Any constant added or subtracted (change of origin) then the standard deviation of original data and of change data after addition or subtraction will not change but the mean of new data will change.
2. Any constant multiplied or divided (change of scale) then mean, standard deviation and variance will change of new changed data.

	Arithmetic Mean	Geometric Mean	Harmonic Mean	Median	Mode
Advantages	• Rigidly defined • Further algebraic treatment possible • All observations considered for calculation • Easy calculation • Least affected by sampling changes	• Rigidly defined • Algebraic treatment possible • All observations considered for calculation • Less affected by outliers or sampling changes	• Gives large weight to small values and lower weight to larger values • All observations considered for calculation	• Easy to calculate and understand • Not affected by outliers • Rigidly defined • Easily calculated even for distributions with open classes	• Easy to calculate, even by observing and understand • Not affected by outliers • Easily calculated even for distributions with open classes
Disadvantages	• Highly affected by outliers / extreme values • Cannot be calculated if even one observation is missing	• Not easy to calculate and interpret • If there is an observation with value zero, the G.M. will be zero	• Not easy to calculate and interpret • If there is an observation with value zero, the H.M. will be zero	• Further algebraic treatment not possible • Highly affected by sampling changes • Not based on all observations	• Algebraic treatment not possible • For some series mode may not be defined or hard to locate (grouped distributions)
Main Uses	• Widely used measure of central tendency • Calculate final grade of students in a class based on multiple subject marks • Average age of cabin crew in flights	• Calculations of index numbers measuring changes in variables • Calculate annual returns on portfolio of stocks	• Useful for situations where we are measuring time, speed, price	• Better measure for assessing salary, net worth, etc. data • Useful for situations where calculations on data is not possible but data can be ordered like IQ, happiness, etc.	• Used for situations like voting • Business forecasting of orders for multi-type products like shirts of different sizes